

SEQUENCE & SERIES

- **Sequence:** A sequence is a function whose domain is the set N of natural numbers and range a subset of real numbers or complex numbers.
- **Real Sequence:** A sequence is a function whose domain is the set N of natural numbers and range a subset of real numbers or complex numbers.
-
- **General term of the Sequence:** The general term of the sequence can be denoted as $\{a_n\}$ or $\{t_n\}$.
-
- **Finite Sequence:** A sequence is called as finite if it has finite number of terms.
- **Infinite Sequence:** A sequence is called as infinite if it has infinite number of terms.
- **Series:** If $\{a_n\}$ represents the sequence then their summation is called as series and is denoted by $\sum_{k=1}^n a_k$ or $\sum_{k=1}^{\infty} a_k$.
- **Finite Series:** A series is called as finite if it the sum of finite number of terms.
- **Infinite Series:** A series is called as infinite if it the sum of infinite number of terms.
- **Progression:** If the terms of the sequence follow certain pattern, then the sequence is called as a progression. There are three types of progressions:
 - (1) Arithmetic Progression.
 - (2) Geometric Progression
 - (3) Harmonic Progression.
- **Arithmetic Progression(AP):** A sequence of numbers is said to be in Arithmetic progression (AP) if the difference between two consecutive terms are same.
Arithmetic progression is of the form: $a, a + d, a + 2d, a + 3d, \dots$
- **nth Term of an Arithmetic Progression:** nth term of an arithmetic progression can be written as $t_n = a + (n-1)d$, where a is the first term and d is the common difference of an arithmetic progression.
 - (1) If an AP has n terms then the last term of AP is $l = a + (n-1)d$.
 - (2) If three numbers a, b, c are in AP then $b - a = c - b \Rightarrow a + c = 2b$.
 - (3) If a is the first term and d is the common difference of an arithmetic progression having m terms, then the n th term from end is the $(m - n + 1)$ th term.
So, n th term from end is $a + (m - n)d$.
 - (4) Three number in AP $a - d, a, a + d$.
 - (5) Four number in AP $a - 3d, a - d, a + d, a + 3d$.
 - (6) Five number in AP $a - 3d, a - d, a, a + d, a + 3d$.

• **Sum of n terms of an Arithmetic Progression:**

$$(1) \quad S_n = \frac{n}{2} \{2a + (n-1)d\}.$$

$$(2) \quad S_n = \frac{n}{2} \{a + l\}.$$

$$(3) \quad t_n = S_n - S_{n-1}.$$

• **Properties of Arithmetic Progression:**

(a) If $a_1, a_2, a_3, \dots, a_n$ are in AP, then:

(1) $a_1 + k, a_2 + k, a_3 + k, \dots, a_n + k$ are in AP.

(2) $a_1 - k, a_2 - k, a_3 - k, \dots, a_n - k$ are in AP.

(3) $a_1 \times k, a_2 \times k, a_3 \times k, \dots, a_n \times k$ are in AP.

(4) $a_1 / k, a_2 / k, a_3 / k, \dots, a_n / k$ are in AP.

(5) $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \dots$

(6) $a_r = \frac{a_{r-k} + a_{r+k}}{2}$, where $0 \leq k \leq n - r$.

(7) If the number of terms of any AP is odd, then sum of the terms is equal to product of middle terms and number of terms.

(8) If the number of terms of any AP is odd then its middle term is A.M. (arithmetic mean) of first and last term.

(b) If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two AP's, then:

(1) $a_1 + b_1, a_2 + b_2, a_3 + b_3, \dots, a_n + b_n$ are in AP.

(2) $a_1 - b_1, a_2 - b_2, a_3 - b_3, \dots, a_n - b_n$ are in AP.

• **Arithmetic Mean:**

(1) **Single Arithmetic Mean:** If A is a single arithmetic mean between two numbers a, b then a, A, b are in AP.

(2) **n Arithmetic Mean:** The numbers $A_1, A_2, A_3, \dots, A_n$ are n arithmetic mean between two numbers a, b , then $a, A_1, A_2, A_3, \dots, A_n, b$ are in AP.

(3) **Inserting Single A.M. between two given numbers:** If A is the arithmetic mean between two numbers a, b then $A = \frac{a+b}{2}$.

(4) **Inserting n A.M. between two given numbers:** If $A_1, A_2, A_3, \dots, A_n$ are n arithmetic mean between two numbers a, b then $d = \frac{b-a}{n+1}$.

$$\text{So, } A_1 = a + \frac{b-a}{n+1}, A_2 = a + 2\left(\frac{b-a}{n+1}\right), A_3 = a + 3\left(\frac{b-a}{n+1}\right), \dots, A_n = a + n\left(\frac{b-a}{n+1}\right)$$

$$\text{Sum of n arithmetic mean } A_1 + A_2 + A_3 + \dots + A_n = n\left(\frac{a+b}{2}\right).$$

- **Geometric Progression(GP):** A sequence of numbers is said to be in Geometric progression (GP) if the ratio between two consecutive terms are same.
Geometric Progression is of the form: $a, ar, ar^2, ar^3, \dots, ar^{n-1}, \dots$.
- **nth Term of an Geometric Progression:** nth term of a geometric progression can be written as $t_n = ar^{n-1}$, where a is the first term and r is the common ratio of a geometric progression.
 - (1) If a is the first term and r is the common ratio, then a GP is of the form $a, ar, ar^2, ar^3, \dots, ar^{n-1}, \dots$
 - (2) If an GP has n terms then the last term of GP is $l = ar^{n-1}$.
 - (3) If l is the last term and r is the common ratio, then nth term from the end is $\frac{l}{r^{n-1}}$.
 - (4) If a is the first term and r is the common ratio of an geometric progression having m terms, then the n th term from end is the $(m - n + 1)$ th term.
So, n th term from end is ar^{m-n} .
 - (5) Three number in GP $\frac{a}{r}, a, ar$.
 - (6) Four numbers in GP $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$.
 - (7) Five numbers in GP $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$.
 - (8) If three numbers a, b, c are in GP, then $\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$.
- **Sum of n terms of a Geometric Progression:**
 - (1) $S_n = \frac{a(r^n - 1)}{r - 1}$ if $r > 1$.
 - (2) $S_n = \frac{a(1 - r^n)}{1 - r}$ if $r < 1$.
 - (3) $S_n = na$ if $r = 1$.
 - (4) If l is last term of a GP, then $S_n = \frac{lr - a}{r - 1}$ if $r \neq 1$.
- **Sum of ∞ terms of a Geometric Progression:**
 - (1) $S_\infty = \frac{a}{1 - r}$ if $|r| < 1$
 - (2) $S_\infty = \infty$ if $|r| \geq 1$.
- **Properties of Geometric Progression:**
 - (a) If $a_1, a_2, a_3, \dots, a_n$ are in GP, then:
 - (1) $a_1k, a_2k, a_3k, \dots, a_nk$ are in GP.
 - (2) $a_1/k, a_2/k, a_3/k, \dots, a_n/k$ are in GP.
 - (3) $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in GP.

- (b) If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two GP's, then:
- (1) $a_1 b_1, a_2 b_2, a_3 b_3, \dots, a_n b_n$ are in GP.
 - (2) $a_1 / b_1, a_2 / b_2, a_3 / b_3, \dots, a_n / b_n$ are in GP.
- (c)
- (1) In a finite GP, the product of terms equidistant from the beginning and the end is always the same and is equal to the product of the first and last term. If $a_1, a_2, a_3, \dots, a_n$ are in GP, then $a_1 a_n = a_2 a_{n-1} = a_3 a_{n-2} = \dots = a_r a_{n-r+1}$.
 - (2) $a_r = \sqrt[n-r+1]{a_1 a_{n-r+2} \dots a_n}$ for $0 \leq k \leq n-r$.
 - (3) If the terms of a given GP are chosen at regular interval. Then the new sequence so formed also forms a GP.
 - (4) If $a_1, a_2, a_3, \dots, a_n$ is a GP of non-zero, non-negative terms then $\log a_1, \log a_2, \log a_3, \dots, \log a_n$ are in AP.
 - (5) If the first term of a GP of n terms is a and the last term is l , then the product of all terms of GP is $(al)^{n/2}$.
 - (6) If there are n terms in a GP, with common ratio r and S_m denotes the sum of first m terms, then sum of their products taken two by two is $\frac{r}{r+1} S_{n-1} S_n$.
 - (7) If $a_1^{x_1}, a_2^{x_2}, a_3^{x_3}, \dots, a_n^{x_n}$ are in GP, then $x_1, x_2, x_3, \dots, x_n$ are in AP.

• **Geometric Mean(GM):**

- (1) **Single Geometric Mean:** If A is a single geometric mean between two numbers a, b then a, A, b are in GP.
- (2) **n Geometric Mean:** The numbers $G_1, G_2, G_3, \dots, G_n$ are n geometric mean between two numbers a, b , then $a, G_1, G_2, G_3, \dots, G_n, b$ are in GP.
- (3) **Inserting Single G.M. between two given numbers:** If G is the geometric mean between two numbers a, b then $G = \sqrt{ab}$.
- (4) **Inserting n G.M. between two given numbers:** If $G_1, G_2, G_3, \dots, G_n$ are n geometric mean between two numbers a, b then $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$.

$$G_1 = a \left(\frac{b}{a}\right)^{\frac{1}{n+1}}, G_2 = a \left(\frac{b}{a}\right)^{\frac{2}{n+1}}, G_3 = a \left(\frac{b}{a}\right)^{\frac{3}{n+1}}, \dots, G_n = a \left(\frac{b}{a}\right)^{\frac{n}{n+1}}.$$
- (5) The product of n geometric mean between two given numbers is the n th power of single GM between them. If $G_1, G_2, G_3, \dots, G_n$ are n geometric mean between two numbers a, b , then $G_1 G_2 G_3 \dots G_n = (\sqrt{ab})^n$.
- (6) If A and G are the arithmetic mean and geometric mean between two numbers a, b then:
 - (i) $A > G$.

- (ii) The quadratic equation having a, b as its roots is $x^2 - 2Ax + G^2 = 0$.
- (iii) The two positive numbers are $A \pm \sqrt{A^2 - G^2}$.

- **Harmonic Progression(HP):** A sequence of non-zero numbers $a_1, a_2, a_3, \dots, a_n$ are said to be in HP if $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP.

- **nth term of Harmonic Progression:** nth term of HP = $\frac{1}{\text{nth term of AP}}$.

- **Harmonic Mean:**

- (1) **Single Harmonic Mean:** If H is a single harmonic mean between two numbers a, b then a, H, b are in HP or $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in AP.

- (2) **n Harmonic Mean:** The numbers $H_1, H_2, H_3, \dots, H_n$ are n harmonic mean between two numbers a, b , then $a, H_1, H_2, H_3, \dots, H_n, b$ are in HP or $\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}, \frac{1}{b}$ are in AP.

- (3) **Inserting Single H.M. between two given numbers:** If H is the harmonic mean between two numbers a, b then $H = \frac{2ab}{a+b}$.

- (4) **Inserting n H.M. between two given numbers:** If $H_1, H_2, H_3, \dots, H_n$ are n harmonic mean between two numbers a, b $a, H_1, H_2, H_3, \dots, H_n, b$ are in HP or $\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}, \frac{1}{b}$ are in AP.

Common difference of AP, $d = \frac{a-b}{ab(n+1)}$.

$$H_1 = \frac{ab(n+1)}{bn+a}, H_2 = \frac{ab(n+1)}{b(n-1)+2a}, H_3 = \frac{ab(n+1)}{b(n-2)+3a}, \dots, H_n = \frac{ab(n+1)}{b+na}$$

- (5) Three terms a, b, c are in HP if $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP or $b = \frac{2ac}{a+c}$.

- (6) No terms of HP can be zero.

- (7) There is no general formula for finding the sum of n terms of HP.

- (8) Reciprocal of terms of HP are in AP, then properties of AP can be used.

- (9) If H is HM between two numbers a, b , then

(i) $\frac{1}{H-a} + \frac{1}{H-b} = \frac{1}{a} + \frac{1}{b}$.

(ii) $(H-2a)(H-2b) = H^2$.

(iii) $\frac{H+a}{H-a} + \frac{H+b}{H-b} = 2$.

- **Relation between AM, GM, HM:** Let A, G and H are the AM, GM and HM between two positive numbers a, b , then:

$$(1) \quad A = \frac{a+b}{2}, G = \sqrt{ab}, H = \frac{2ab}{a+b}.$$

$$(2) \quad A \geq G \geq H \text{ and equality holds if } a = b.$$

$$(3) \quad \frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \begin{cases} A & \text{if } n = 0 \\ G & \text{if } n = -\frac{1}{2} \\ H & \text{if } n = -1 \end{cases}.$$

$$(4) \quad \text{If } A_1, A_2 \text{ are two AM's, } G_1, G_2 \text{ are two GM's and } H_1, H_2 \text{ are two HM's between two numbers } a, b, \text{ then } \frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}.$$

- **Relation between AP, GP, HP:**

$$(1) \quad \text{If } \frac{a-b}{b-c} = \frac{a}{a}, \text{ then } a, b, c \text{ are in AP.}$$

$$(2) \quad \text{If } \frac{a-b}{b-c} = \frac{a}{b}, \text{ then } a, b, c \text{ are in GP.}$$

$$(3) \quad \text{If } \frac{a-b}{b-c} = \frac{a}{c}, \text{ then } a, b, c \text{ are in HP.}$$

- **Some Special Sequence:**

$$(1) \quad 1 + 2 + 3 + \dots + n = \sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

$$(2) \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$(3) \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

$$(4) \quad 1^4 + 2^4 + 3^4 + \dots + n^4 = \sum_{k=1}^n k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

- **General rule to find the value of recurring decimal:** Let X denotes the figure, which do not recur and assume they are l in numbers. Let Y denote recurring period of consisting of m figures. Let R denotes the value of recurring decimal. Let $R = 0.XYYY\dots$

$$\Rightarrow 10^l R = X.YYY\dots$$

$$\Rightarrow 10^{l+m} R = XY.YYY\dots$$

$$\Rightarrow (10^{l+m} - 10^l)R = XY - X$$

$$\Rightarrow R = \frac{XY - X}{10^{l+m} - 10^l}.$$

- **Arithmetico-Geometric Sequence (AGS):** A sequence is said to be arithmetico-geometric sequence if each term is the product of corresponding terms of AP and GP i.e. if $a_1, a_2, a_3, \dots, a_n$ are in AP and $b_1, b_2, b_3, \dots, b_n$ are in GP, then $a_1b_1, a_2b_2, a_3b_3, \dots, a_nb_n$ are in AGP.

An arithmetico-geometric sequence is of the form:

$$ab, (a+d)br, (a+2d)br^2, (a+3d)br^3, \dots, (a+(n-1)d)br^{n-1}, \dots$$

- **nth Term of Arithmetico-Geometric Sequence:** Let an arithmetico-geometric sequence is of the form $ab, (a+d)br, (a+2d)br^2, (a+3d)br^3, \dots, (a+(n-1)d)br^{n-1}, \dots$ then its nth term can be denoted as $t_n = (a+(n-1)d)br^{n-1}$.
- **Sum of terms of Arithmetico-Geometric Sequence:**

$$(1) \quad S_n = \frac{ab}{1-r} + \frac{bdr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]br^n}{1-r}.$$

$$(2) \quad S_\infty = \frac{ab}{1-r} + \frac{bdr}{(1-r)^2}.$$

$$(3) \quad \text{If } r \geq 1, \text{ then } S_\infty \text{ does not exist.}$$

- **Method for Finding Sum (Method of Difference):** It is applicable for both sum of finite and infinite terms.

Let Sum of the series is S, then multiply it by the common ratio of GP and subtract, which yields a geometric series whose sum can be easily obtained.

If the difference of the successive terms of a series in Arithmetic series of Geometric Series we can find nth term of the series by the following:

- (1) Denote nth term by T_n and the sum of n terms of the series as S_n .
- (2) Rewrite the given series with each term shifted by one place to right.
- (3) By subtracting the later series from the first series, find T_n
- (4) From T_n , S_n can be found by appropriate summation.

- **Lagrange's Identity:**

$$\begin{aligned} & (x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2)(y_1^2 + y_2^2 + y_3^2 + \dots + y_n^2) - (x_1y_1 + x_2y_2 + x_3y_3 + \dots + x_ny_n)^2 \\ &= (x_1y_2 - x_2y_1)^2 + (x_1y_3 - x_3y_1)^2 + (x_1y_4 - x_4y_1)^2 + \dots + (x_1y_n - x_ny_1)^2 \\ & \quad + (x_2y_3 - x_3y_2)^2 + (x_2y_4 - x_4y_2)^2 + \dots + (x_2y_n - x_ny_2)^2 \\ & \quad + (x_3y_4 - x_4y_3)^2 + \dots + (x_3y_n - x_ny_3)^2 \\ & \quad \dots + (x_{n-1}y_n - x_ny_{n-1})^2 \end{aligned}$$
